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Lehrbuch der Algebra
Working English Translation Batch 08
Transposed Substitutions, Fractional Linear Actions, and
Permutation Groups

Translator's note

This working translation continues the Weber line from the start of linear substitutions. Batch 07 translated the definition of linear substitutions, their composition law, and inverse substitutions. The present batch continues with transformed and transposed substitutions, then translates the next two sections on fractional linear substitutions in one variable and permutation groups as linear substitution groups. I have preserved Weber's order of exposition and formulas while lightly regularizing punctuation and notation.

Continuation of §37. Transformed and transposed substitutions

If a set of substitutions has the property that the composition of any two of its elements again gives an element of the same set, then that set likewise forms a group, which may be finite or infinite.

Let L be a fixed substitution. From every substitution A of the same dimension one may derive a substitution

$$A' = L^{-1}AL \quad (21)$$

called the *transform* of A by L . If we set

$$(x) = L(x'), \quad (y) = L(y'), \quad (22)$$

then from the defining relation for A it follows that

$$(y') = A'(x'), \quad (23)$$

so that passing to the transformed substitution is equivalent to applying the same change of variables L^{-1} simultaneously to both systems of variables.

If

$$A' = L^{-1}AL, \quad B' = L^{-1}BL,$$

then the composition laws give

$$A'B' = L^{-1}ABL,$$

and therefore:

7. As A runs through the substitutions of a group, the transformed substitutions $L^{-1}AL$, for fixed L , run through the substitutions of an isomorphic group.

The inverse substitutions are to be distinguished carefully from the transposed substitutions.

From each substitution A one obtains another specific substitution, called the *transposed substitution*, which for the moment we denote by A^t , by interchanging rows and columns in A - that is, by swapping the upper and lower indices of the coefficients.

If in the composition formula one interchanges lower and upper indices and at the same time exchanges the roles of a and b , one obtains the rule

$$(BA)^t = A^t B^t. \quad (24)$$

Thus the transpose behaves under composition in a way closely analogous to the inverse substitution: in taking transposes one must reverse the order of the factors, whereas for inverses one preserves the order and replaces each factor by its inverse. Combining the two rules gives

$$(A^{-1})^t = (A^t)^{-1}, \quad (25)$$

which Weber abbreviates, when no ambiguity is possible, by A_t^{-1} .

§38. The group of integral linear substitutions of one variable

We now apply the general results of the preceding paragraph to several special cases and begin with substitutions of one variable. These substitutions are identical with integral fractional-linear functions of the first degree:

$$y = \frac{\alpha x + \beta}{\gamma x + \delta}, \quad (26)$$

for which we introduce, for brevity, the symbol

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \quad (27)$$

and whose composition is governed by the formula

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} = \begin{pmatrix} \alpha\alpha' + \beta\gamma' & \alpha\beta' + \beta\delta' \\ \gamma\alpha' + \delta\gamma' & \gamma\beta' + \delta\delta' \end{pmatrix}. \quad (28)$$

The coefficients $\alpha, \beta, \gamma, \delta$ are arbitrary numbers whose determinant $\alpha\delta - \beta\gamma$ is nonzero.

For the identity substitution one has

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad (29)$$

and the inverse substitution of (26) is

$$\frac{1}{D} \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix}, \quad D = \alpha\delta - \beta\gamma. \quad (30)$$

The transposed substitution attached to (26) is obtained by interchanging the diagonal entries and simultaneously changing the signs of the off-diagonal entries:

$$\begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix}, \quad (31)$$

or, what is the same, by replacing $\alpha, \beta, \gamma, \delta$ by $\delta, -\gamma, -\beta, \alpha$. One verifies immediately that

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix} = \begin{pmatrix} D & 0 \\ 0 & D \end{pmatrix}, \quad (32)$$

which on the one hand confirms formula (24), and on the other shows that every such substitution commutes with its transpose whenever D is a symmetric function of α and δ .

§39. Permutation groups as linear substitution groups

Let $x_1, x_2, \dots, x_n$ be a system of n variables, and let $\alpha_1, \alpha_2, \dots, \alpha_n$ be some ordering of the digits $1, 2, \dots, n$. Then the equations

$$x_1 = x'_{\alpha_1}, \quad x_2 = x'_{\alpha_2}, \quad \dots, \quad x_n = x'_{\alpha_n} \quad (33)$$

determine a linear substitution of dimension n ,

$$A = (a_i^{(k)}), \quad (x) = A(x'), \quad (34)$$

in which each row and each column contains exactly one nonzero coefficient, and that coefficient is equal to 1.

The determinant $|A|$ is therefore equal to ± 1 . If, after carrying out the substitution, one sets $x'_i = x_i$ again, the result is simply the permutation

$$\begin{pmatrix} 1, 2, \dots, n \\ \alpha_1, \alpha_2, \dots, \alpha_n \end{pmatrix}$$

of the indices of the variables.

If B is a second substitution of the same kind,

$$(x') = B(x''), \quad (35)$$

or explicitly

$$x'_1 = x''_{\beta_1}, \quad x'_2 = x''_{\beta_2}, \quad \dots, \quad x'_n = x''_{\beta_n}, \quad (36)$$

then the composition rules of §37 yield

$$x_1 = x''_{\beta_{\alpha_1}}, \quad x_2 = x''_{\beta_{\alpha_2}}, \quad \dots, \quad x_n = x''_{\beta_{\alpha_n}}, \quad (37)$$

which is abbreviated by

$$(x) = AB(x''). \quad (38)$$

But according to the composition law for permutations one has

$$\begin{pmatrix} 1, 2, \dots, n \\ \alpha_1, \alpha_2, \dots, \alpha_n \end{pmatrix} \begin{pmatrix} 1, 2, \dots, n \\ \beta_1, \beta_2, \dots, \beta_n \end{pmatrix} = \begin{pmatrix} 1, 2, \dots, n \\ \beta_{\alpha_1}, \beta_{\alpha_2}, \dots, \beta_{\alpha_n} \end{pmatrix}. \quad (39)$$

Thus, if the substitutions A and B are regarded as permutations of the indices, then the substitution AB is exactly the composed permutation AB .

Hence permutation groups on n symbols are nothing other than a special case of finite groups of linear substitutions of dimension n .

Permutations of the first kind correspond to substitutions with determinant $+1$, and permutations of the second kind correspond to substitutions with determinant -1 .

This relation between permutations and linear substitutions enables us to derive from every permutation group a group of linear substitutions, and conversely to settle certain questions about permutation groups with the aid of the theory of linear substitutions.